SIMATS Online Programming Lab

C PROGRAMMING

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CO-1 Assessment Tool -3 Pseudo code

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QUESTIONS

Q1
Pseudo Code writing, input, output and description for the following problems required
100 marks

1/1 answered

Q1 **Pseudo Code writing, input, output and description for the following problems required** 100 marks

Q1. Given pseudocode:
#define DEBUG 0

IF DEBUG
PRINT intermediate_value
END IF

- Identify error in directive usage
- Correct the logic

Q2. #define PI ______

DECLARE radius AS float
DECLARE area AS float

INPUT radius

area = ______

PRINT area
Fill:

- Constant
- Formula

Q3. #define RATE 1.5

DECLARE x AS int = 10
DECLARE result AS float

result = x * RATE

PRINT result
Questions:

- Output?
- Why is the result float?

Q4. #define MAX 1000

DECLARE temp AS int
INPUT temp

IF temp > MAX
PRINT "Valid"
ELSE
PRINT "Invalid"

- Identify logical error
- Suggest correct condition

Q5. Given unordered pseudocode steps:

- DEFINE conversion rate
- PRINT result
- INPUT amount
- CALCULATE converted value

Arrange in correct order

Q6. #define MIN 10
#define MAX 50

DECLARE value AS int
INPUT value

_____ (value >= MIN AND value <= MAX)
PRINT "Within range"

_____
PRINT "Out of range"
Fill missing control structures

Q7. A billing system includes optional features such as discount calculation and tax computation. Depending on the business requirement, these features may be enabled or disabled at compile time. The program should calculate the total bill amount based on input price and quantity. If discount and tax features are enabled, they should be applied using predefined constants; otherwise, they should be skipped. The system must ensure that floating-point precision is maintained during calculations.

Write pseudocode using preprocessor directives to enable/disable features, define constants for discount and tax rates, and use appropriate data types.

Q8. #define TAX 5%

DECLARE price AS int
DECLARE total AS int

total = price + (price * TAX)

PRINT total

- Identify errors
- Correct data types and constant definition

9. A temperature conversion program supports multiple modes such as Celsius to Fahrenheit and Fahrenheit to Celsius. The mode of operation should be selected at compile time using preprocessor directives. The program should use appropriate floating-point data types to ensure accuracy and define conversion formulas using symbolic constants wherever applicable. The final output should be displayed with two decimal precision.
Write pseudocode using preprocessor directives for mode selection, constants for conversion formulas, and proper data types.

10. #define SIZE 5

DECLARE arr[SIZE]
INPUT elements

- Extend pseudocode to:
  - ○ Find sum
  - ○ Compute average

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